

Local Communication Control for Pre-Disaster Monitoring with Intermittent Backhaul

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Abstract—Intermittent backhaul disconnects a monitoring controller without stopping the communication state it has installed. Unacknowledged records continue consuming access opportunities, and applied sampling profiles continue drawing energy after corrective commands become unreachable. We study this failure in pre-disaster mountain monitoring with battery/solar LoRaWAN Class A nodes, a cellular primary and an uplink-only short-message backup. We formulate an execution contract that separates record lifetime, authorised configuration fallback and evidence of end-to-end completion, and examine where each decision can be enforced. The central observation is that a valid local action can require less information than complete failure diagnosis. In the tested cache model, placing standard deadline expiry at the source improves on-time service by 4.04 percentage points over ten paired seeds and increases outage on-time delivery by 2.45 $\times$; gateway-only suppression can instead back-pressure access. Node-local protection also prevents losses that a central downgrade cannot avert in the tested episodes. Ordinary TTL and battery rules cover the evaluated configuration-stopping problem, including a same-information exact reference, so a new lease algorithm is not required for these results. Partial gateway attribution and live Agent traces expose a separate interface failure: access receipts being interpreted as backhaul or completion evidence. These findings support local enforcement and explicit evidence boundaries as communication-system design principles. The Agent study is formative; an independent benefit of the complete execution interface remains to be evaluated against a same-component ordinary stack.

Index Terms—Pre-disaster monitoring, intermittent backhaul, LoRaWAN Class A, communication state, local enforcement, deadline expiry, partial observability, agent execution.

I. INTRODUCTION

A mountain monitoring node receives an authorised dense sampling profile while its backhaul is available. The cellular path then fails, but the node continues sampling and transmitting. A later downgrade cannot reach its Class A receive window. Meanwhile, records lacking central acknowledgement remain in the source cache and occupy later access batches even after their business deadlines. The gateway may suppress useless return traffic without releasing the source copies. Correct decisions at one layer can therefore fail to stop resource consumption at another.

This paper studies *persistent communication state under disconnected control*. Record retention and applied configuration have different termination conditions, but share an execution dependency: waiting for a later central decision assumes a control path that may be unavailable. The monitoring obligation remains end-to-end—an in-window sample must

arrive at the centre before its deadline—while evidence and authority are distributed across the node, gateway and centre. An access receipt confirms one stage of that chain, not the entire obligation.

A. Design insight

Local action and complete diagnosis have different information requirements. A gateway may be unable to distinguish a sample that was never collected from one not yet heard. The source nevertheless knows whether a stored record has passed its locally assigned deadline and can stop retransmitting it. Similarly, an authorised local fallback can act on a clock or battery condition even when the centre cannot determine the current profile. These actions need explicit authority and enforceable local predicates; they do not need an invented account of the remote failure.

We organise this distinction as an execution contract: identify the state an operation changes, the evidence supporting its outcome, and the location capable of acting before further harm. Standard expiry and local protection instantiate the contract. The contribution is evaluated through their placement and communication consequences, rather than through a new name for these established mechanisms.

B. Ordinary mechanisms and the remaining question

The comparisons include standard per-record expiry, local clock and battery protection, fixed-duration fallback, strong backup packing and a same-information exact stopping reference. Expiry reproduces obligation-keyed deletion on the tested seeds, and ordinary configuration rules cover the evaluated stopping family. These results delimit the need for a new algorithm. They do not establish a global optimum for communication scheduling.

A remaining systems question is whether an explicit execution interface improves action selection and repair beyond a stack already containing those ordinary components. The distinction matters for Agents: a planner may order more sampling when the missing stage is return delivery, or infer that a command applied from unrelated access receipts. Our live study observes the latter interface failure, but its end-to-end comparison does not yet isolate the interface’s independent contribution. We keep that question separate from the deterministic placement results.

C. Contributions

- 1) **An execution contract for disconnected monitoring** (§III–§V). The formulation separates locally decidable record expiry, authorised configuration fallback and partial end-to-end evidence. An enforcement condition combines evidence availability, authority and actuation delay.
- 2) **Measured cross-segment placement effects** (§VII-B, §VII-E). Source-local standard expiry improves service by 4.04 percentage points (95% CI [3.48,4.60]) and outage on-time delivery by $2.45\times$. A complementary configuration experiment shows why a central protection rule can fail when the same rule can still act locally.
- 3) **An empirical boundary for control and evidence** (§VII-C–§VII-F). Same-information ordinary stopping rules match the declared exact reference; gateway attribution remains necessarily partial; and formative Agent traces identify access/backhaul evidence confusion. Together these comparisons separate established local remedies from unproven planner or runtime increments.

The setting is pre-disaster monitoring with externally authorised requirements. The system does not infer hazard levels, remove missed obligations or obtain additional communication resources by changing the scoring task.

II. SCENARIO AND OPERATING POINT

Fourteen sensor nodes in two groups monitor displacement and rainfall on a slope. Nodes sample and upload to a local gateway over LoRaWAN Class A; the gateway forwards to a central server over cellular backhaul, with a BeiDou short-message service as a sparse, metered backup carrying field-to-centre messages only, never downlink configuration. Nodes are battery plus small solar, the mission is pre-disaster and routine with authorised warning-level changes, and no component judges landslide risk. Node count, role mapping and disturbance parameters follow the frozen Task v1.1 instance manifest; an earlier 16-node sketch is not used as a field fact.

Authority comes from two Chinese geological-hazard monitoring standards, DZ/T 0460–2023 and DZ/T 0450–2023, retrieved and clause-audited in the repository [1], [2]. §5.3.3 permits remote dynamic adjustment of sampling and upload frequency by warning level and gives no numeric levels, so the periods used here are declared research choices; §8.4.1 defines four levels and §8.4.2 permits upgrade, downgrade or termination after a conference; §5.3.7 calls for dual-mode communication under weak coverage in high-risk extreme weather—a conditional clause appearing once, not a blanket dual-channel mandate, and the communication-layer standard does not impose it. The BeiDou open-service specification (BDS-OS-PS-3.0) [3] reports $\geq 95\%$ success, sub-2 s average latency, an average service frequency of one message per 30 s and a 14,000-bit single-message ceiling, noting that realised frequency and maximum length are constrained by registration parameters: the rate is a provisioned quantity. These clauses bound lawful action—re-compile frequency within granted levels and request a conference change, never silently drop an obligation or declare risk.

The backup uses one 78-byte packet per 1200 s (20 B header, 58 B payload; a displacement sample is 6 B and rainfall 4 B, so roughly 9–10 samples fit). The elevated (yellow) task uses a 300 s period with a delivery deadline two periods after window open, a ≈ 600 s delivery window—shorter than the 1200 s backup slot spacing. The 200 B ceiling and ≥ 1 min period in DZ/T 0450 (§7.4.3.2, §7.4.2.3, with a ≥ 5 s inter-message gap at §7.4.3.5) belong to different service-formulation sections and are not combined into one “BeiDou operating point”; the 1200 s/78 B point is a declared research choice.

III. SYSTEM MODEL

Time is slotted at $\Delta=60$ s and $t=0$ is 06:00 local, so $t=12$ h is sunset and the solar cycle is explicit. Nodes $\mathcal{N}$, measurands $\mathcal{M}$ and a gateway g compose the field; a centre c lies beyond the backhaul.

A. Obligations and the delivery chain

A monitoring obligation is $o = (n, m, [l_o, h_o], d_o)$ over node n , measurand m , sampling window and absolute delivery deadline. A routine obligation at period P has $h_o - l_o = P$ and $d_o = h_o + P$, one period of grace. A sample fulfils o only if it matches (n, m) , is taken in-window and is received centrally by d_o , traversing

$$\underbrace{\text{qualified sample}}_{\text{node}} \rightarrow \underbrace{\text{heard at } g}_{\text{LoRa access}} \rightarrow \underbrace{\text{returned by } d_o}_{\text{cellular / short message}} \rightarrow \underbrace{\text{central evidence}}_{\text{fulfilled}}. \quad (1)$$

For piecewise missions the first segment uses a closed interval and later segments half-open windows, so a boundary sample cannot double-cover adjacent windows after a level change.

B. Links, outages, and who can act

LoRa access (node $\rightarrow$ gateway, with Class A downlink windows) succeeds per opportunity with probability $p_a=0.74$; the primary backhaul (gateway $\rightarrow$ centre) is good with probability $p_b=0.62$ outside a declared outage. A control outage is an interval $\mathcal{O} = [a, b)$ during which the primary is unavailable to centre $\rightarrow$ field traffic: centre commands and mission-table segments are refused at admission and never transmitted, and the short-message backup cannot carry downlink configuration. During $\mathcal{O}$ the only entities able to change field communication behaviour are the *node* (on its clock, own battery, cache, recent access receipts and bounds carried by already-applied commands) and the *gateway* (on samples it has actually heard, its forwarding log and backup slots used). The centre retains authority to authorise changes but no channel to install them; a revision authored in $\mathcal{O}$ reaches the field at b .

C. Persistent installed state

Record state. Each unacknowledged sample s at node n has creation time t_s , a business deadline $d(s) = (\lfloor t_s/P \rfloor + 2)P$ under the locally known cadence, and occupies one of $K=240$ cache slots and up to $B=32$ positions in an access batch. Under the modelled edge discipline it leaves the cache only

on central acknowledgement (modelled synchronously without airtime, equally favouring every queue rule); until then it is re-sent oldest-first.

Configuration state. The node runs one profile $q(t) \in \{P_0, P_1\}$ (sparse $P_0=600$ s, dense $P_1=300$ s) for both sampling and reporting, installed by an applied command carrying a version. A command is *applied* when the node receives and executes it; a later report can confirm that effect to the observer. Queued or in-transit commands change nothing. While applied, $q=P_1$ draws additional sampling and uplink energy every tick until something terminates it.

D. Energy

A sample costs $e_s=4.7 \times 10^{-4}$ Wh and an uplink $e_u=2.33 \times 10^{-5}$ Wh; battery capacity is $C=0.05$ Wh. Solar input is a 12 h half-sine with peak η (nominally 0.03 Wh/h) plus per-node/hour stochastic cloud occlusion. A node whose state of charge reaches zero *permanently* dies in the main configuration. Holding dense over a 12 h night draws ≈ 0.071 Wh $> C$.

E. Local predicates and authorised fallback

A record carries an expiry time $d_n(s)$ assigned from the cadence known at its source. Once $t > d_n(s)$, retaining it has no value for obligations represented by that local contract. Records remain eligible on the deadline tick. During an unreceived mission revision, the source cannot substitute a future central deadline for $d_n(s)$; the independent scorer retains the actual authorised obligations.

A configuration instead has a permitted fallback set and a local trigger, such as an installed TTL, a clock rule or a battery condition. Stopping dense acquisition and deleting previously collected records are separate actions: a sample can remain useful after the profile changes. Consequently we do not assign configurations a universal release time derived from the last return slot. Unknown future mission revisions cannot become node-local stopping evidence.

F. Service loss and statement discipline

The objective counts deadline-valid obligation delivery (service), permanent node deaths, and the resource cost of state outliving its release time: wasted samples, access attempts, short-message bytes and energy. Offline, each unfulfilled obligation is charged to its *first* failing stage under an optimistic model granting every later assumption, giving four mutually exclusive segments: S_{energy} (no qualified sample), S_{access} (sampled but not heard at the gateway by the deadline), S_{time} (heard but no return opportunity exists) and S_{cap} (an opportunity exists but capacity or congestion cannot carry it). An online observer that cannot see a stage reports *unknown* and never guesses it. Every feasibility claim is **confirmed** (an independent report shows the target config, or a matching sample is received centrally), **unconfirmed** (a command is in flight or receipts are incomplete, which implies neither that it took effect nor that the channel is down) or **unreachable** (the current path is known down from explicit network-side state).

A LoRa access receipt—“a node was heard”—is evidence about the access link only and never by itself establishes backhaul reachability.

IV. PROBLEM FORMULATION

Let $\mathcal{H}_e(t)$ contain only the evidence actually available to entity e by time t . Node history includes clock, battery, cache and received commands; gateway history additionally includes heard samples and its own forwarding log. A policy chooses record retention/release and configuration hold/fallback through $a_e(t) = \pi_e(\mathcal{H}_e(t))$. We compare the outcome vector

$$\mathcal{J}(\pi) = (N_{\text{miss}}, N_{\text{brownout}}, E_{\text{node}}, A_{\text{up}}, A_{\text{down}}, B_{\text{backup}}), \quad (2)$$

where missed obligations, brownouts, node energy, uplink/downlink airtime and backup bytes are reported separately. A scalar death penalty is a declared experimental preference in the single-node reference, not an inferred requirement of disaster monitoring. Strict survival is one possible risk regime, not a universal assumption imposed on every experiment.

A. Execution contract

The comparison obeys four conditions. *Locality*: identical local histories cannot produce different actions solely because of an unseen future. *Authority*: configuration fallbacks must lie within permissions granted before they are used. *Retention accountability*: the source may expire a record under its known lifetime; releasing transmission state does not erase an obligation from the score. *Outcome evidence*: a queued request, an applied configuration, a gateway receipt and central delivery establish different facts.

These conditions leave a real tradeoff. A fallback may protect energy while reducing acquisition for an outstanding obligation, and that loss remains in N_{miss} . The system cannot promise both full service and survival under every resource history merely by changing the controller.

B. Where enforcement can act

For a predicate c and candidate layer e , the design condition is

$$\begin{aligned} &\text{evidence}(e, c) \wedge \text{authority}(e, c) \\ &\wedge D_{\text{observe+decide+act}}(e, c) < M_c(t), \end{aligned} \quad (3)$$

where $M_c(t)$ is the remaining intervention margin. Evidence and authority alone do not make a central protection rule enforceable: its delay includes backhaul delivery and a Class A window. During a control outage that delay may exceed the energy margin. A source-local expiry needs neither a new command nor a remote failure diagnosis. The condition identifies implementation requirements; an empirical delay estimate is not a universal safety bound.

C. Knowledge boundary

Consider two histories with identical observations at the node, one with an unseen central downgrade and one with the elevated requirement still active. Any causal node policy takes the same action in both until new evidence arrives. It therefore cannot claim to track the unseen mission revision exactly. A pre-authorised fallback resolves what the node is allowed to do; it does not reveal which remote history occurred. This elementary indistinguishability argument explains why an unknown outcome must remain distinct from confirmed completion or infeasibility.

An Agent, a deterministic rule and a solver can all consume this contract. Their common action space and evidence boundaries define the communication problem independently of the planner.

V. MECHANISMS

The prototype instantiates the execution contract at three locations. The source applies known record lifetimes and authorised fallback rules; the gateway uses heard samples and finite return opportunities; the centre compiles requirements and interprets received evidence. The experiments below isolate these components. Their complete composition is not assumed to outperform an ordinary stack containing the same mechanisms.

A. Record termination: deadline-aligned expiry

At each upload opportunity the node deletes cached records with $d_n(s) < t$ and sends survivors oldest-first. We implement two formulations: `deadline-purge`, keyed to the obligation cadence, and `generic-expiry`, an ordinary BPv7-style per-record lifetime assigned at creation, $\text{expires} = t_s + ((P - t_s \bmod P) + P) = d(s)$, referencing no obligation id, slot geometry or certificate. A gateway variant locally acknowledges records it will not send. EDF and obligation-greedy queues only reorder and never release; AoI `latest-only` releases indiscriminately, keeping only the newest. Dropping stale or expired traffic is well established in AoI packet management and DTN bundle expiry [4], [5], [6], [7], and suppressing sensor traffic whose delay bound is exceeded is likewise prior art [8]. We claim no new discard algorithm. The system-level content is the end-to-end business-deadline basis—not packet age, hop TTL or an estimated single-packet delay—zero-downlink source placement, and, decisively, cross-segment consistency, because the acknowledgement coupling means a gateway-only release does not free the source cache.

B. Configuration fallback with ordinary protection

The local clock guard clamps sampling and reporting to sparse at night and leaves daytime commands untouched. The configuration comparison additionally evaluates fixed TTLs, a rolling battery/expected-harvest check and an energy-derived stopping bound. TTL starts at the actual over-the-air dense-application edge. All compared configurations use the same source expiry, backup packing and permitted fallback profile.

Two information conditions are separated. Under announced validity, the node receives a known validity end with the configuration. Under runtime authorisation, a downgrade issued during the outage remains unknown in the field; local policies use their clock, measured battery and declared energy model. Announced validity is an informative comparator, not a same-information competitor to the latter policies. A separate quantised single-node stopping study compares ordinary rules with an exact non-prescient reference under a public task table. Its scope and findings are reported in §VII-C.

C. Partial-observability projector

At compile time the centre holds the obligation table, the *current* and never forecast network-side outage state, and public parameters; at mission-table arrival the gateway holds only what it has heard. A structural no-return certificate (S_{time}) is emitted from slot geometry alone and is explicitly conditional on the primary being unavailable; geometry never infers an outage. An arrival-time projector labels a missed obligation only when local evidence separates the stages, and otherwise returns unknown. An unknown-aware centre projector emits *unreachable* solely from an explicit network-side field—which the current harness does not yet provide, so it emits *unconfirmed*—and replaces any blanket night kill with the computed energy ledger, so genuine energy-infeasibility is distinguished from conservative advice.

VI. PROTOTYPE AND METHODOLOGY

All mechanisms run in one Python event-stepped simulator (60 s ticks) implementing LoRaWAN Class A downlink windows, versioned two-field configuration commands (sampling interval and reporting period delivered as independent fields, with cross-generation mixed configs detected), the hold-until-acknowledged edge cache, max-coverage backup packing, stochastic solar and cloud, absorbing brownout, and the non-prescient piecewise mission-view gate. Disabling the backup reproduces the v1.1 simulator bit-for-bit across five anchored invariants; the local guard and fallback extensions are default-off.

Two controlled episodes are used. *Episode I*, the v1.1 manifest point, upgrades to yellow at $t=6$ h with primary outage 4–20 h and the mission table reaching the gateway at 20 h; it drives the frontier, attribution, record-expiry, placement and live-agent results. *Episode II* is a deterministic configuration-termination scenario with identical parameters except schedule, outage phasing and swept harvest: to place a daytime downgrade *inside* an outage, phase A uses yellow at $t=2$ h with a conferred downgrade at $t=6$ h and outage 4–20 h, phase B uses yellow at $t=1$ h, downgrade at $t=8$ h and outage 6–22 h; $t=12$ h is sunset. Episode II is an explicitly constructed controlled comparison for the configuration subproblem, not a new field claim. Node deaths are reported as totals over seeds 0–2; sub-point mean-service differences at $n=3$ are treated as not significant; seeds 6–9 are a reused held-out confirmation set, not an untouched test set. Tables are generated from registered result files, and the artifact provides deterministic checks and frozen reference results.

TABLE I
COUNTERFACTUAL INTERVENTIONS OVER 7560 OBLIGATIONS
(EPISODE I). SET DECOMPOSITION OVER THE TESTED CONFIGURATIONS;
NOT AN UPPER BOUND OVER ALL LEGAL POLICIES.

Run	Sampling/Energy	Backhaul	Delivered
base	dayfeed, 0.05 Wh	1200 s/78 B	3025 (0.400)
O_{bh}	dayfeed, 0.05 Wh	unlimited	4740 (0.627)
O_{en}	always-300, 0.50 Wh	1200 s/78 B	4089 (0.541)
O_{bo}	always-300, 0.50 Wh	unlimited	6023 (0.797)

VII. EVALUATION

A. Where the fixed-resource loss is

Over Episode I's 7560 routine obligations the strong maxcov/FIFO baseline delivers 3025 (0.400). Symmetric per-obligation counterfactuals widen one physical resource at a time, never the policy, and report set differences over identical obligation ids. Relaxing backhaul to unlimited gives 4740 (0.627), relaxing energy (0.50 Wh, always-dense) gives 4089 (0.541), and both gives 6023 (0.797); 1715 obligations are blocked by backhaul capacity alone, 1128 by battery alone, and 155 (2.0%) by both (Table I). These are intervention results on implemented strategies, not an exhaustive optimum; we make no zero-remainder or global-optimality claim. They locate substantial resource pressure at this operating point. The state-termination experiments address an additional execution effect under fixed resources; their motivation does not require proving that every scheduler has exhausted its headroom.

Charging each of the 4535 failures to its first failing segment, with "heard" required by the deadline rather than at any later time, gives S_{energy} 2002 (44.1%), S_{cap} 1115 (24.6%), S_{access} 830 (18.3%) and S_{time} 588 (13.0%). An earlier heard-ever accounting had reported S_{cap} 1945 and S_{access} 0: all 830 moved obligations were in fact sampled but first heard at the gateway after their deadline, late access rather than backhaul capacity. The delivered count (3025) is identical under both accountings; the correction changes where the loss is charged, and therefore which mechanism could address it.

B. Record expiry: standard primitive, decisive placement

During the Episode I outage 75 backup packets carry 750 records; under FIFO 219 arrive centrally already expired against 261 on-time (45.6%; 1270 of 2782 payload bytes). Gateway-only suppression removes those sends but withholds the acknowledgement, and on-time delivery falls 202 $\rightarrow$ 194: cost moves to the access segment. Source-local deadline expiry reverses the sign, raising outage on-time delivery 202 $\rightarrow$ 420 Across ten seeds (Table II) expiry raises mean service 0.373 $\rightarrow$ 0.413, a paired +4.04 points (95% CI +3.48 to +4.60, positive on all ten), total outage on-time delivery 1691 $\rightarrow$ 4145 (2.45 $\times$), expired backup records 2538 $\rightarrow$ 120, and deaths 12 $\rightarrow$ 0; it beats AoI latest-only by +5.55 points, because latest-only buys outage freshness by abandoning post-recovery coverage. Crucially, deadline-purge and generic-expiry are bit-identical on every seed—service, outage counts, deaths and delivered set—so the record mechanism is standard per-record expiration, and its measured

TABLE II
TEN-SEED EPISODE I SWEEP WITH THE STRONG MAXCOV CHOOSER
FIXED; ONLY THE EDGE DISPOSITION CHANGES. GATEWAY-ONLY
SUPPRESSION (NOT SHOWN; SAME FIFO CACHE) GIVES 194 OUTAGE
ON-TIME AT SEED 0, AGAINST 202FOR FIFO AND 420FOR SOURCE
EXPIRY—THE CROSS-SEGMENT COST. $\dagger$ STANDARD BPV7-STYLE
PER-RECORD LIFETIME; $\ddagger$ OBLIGATION-KEYED PURGE, BIT-IDENTICAL TO
 $\dagger$.

Edge disposition	Full svc	Outage OT	Expired	Deaths
FIFO (baseline)	0.373	1691	2538	12
latest-only (AoI)	0.358	4250	0	0
generic-expiry $\dagger$	0.413	4145	120	0
deadline-purge $\ddagger$	0.413	4145	120	0

value is the business-deadline basis plus source/gateway cross-segment consistency, not a new discard algorithm.

C. Configuration protection and its algorithmic boundary

Episode II holds record expiry and maxcov packing fixed while comparing local fallback rules. The clock-only rule reacts at sunset; fixed-duration and energy rules can act earlier. The announced-validity row has advance mission-end information and is shown separately from policies that cannot observe a downgrade during the outage.

At $\eta = 0.012$, the clock guard leaves 20 and 31 deaths in phases A and B. Fixed TTL 8 h yields zero deaths and yellow delivery counts 849/2016 and 2020/3528, whereas TTL 4 h loses 104 deliveries in phase B. The evaluated energy-derived gate also yields zero deaths but loses 290 and 826 yellow deliveries relative to TTL 8 h. Its predictive load approximation is conservative relative to the measured radio load; these results characterise that implementation, not all predictive energy control. The matrix establishes the utility of local fallback while providing no independent increment for the proposed energy-derived lease.

Same-information stopping reference. To separate an implementation limitation from optimisation headroom, a quantised single-node model evaluates continued dense operation versus an absorbing return to sparse over the full physical horizon. Its public task table determines reward; cloud realisations remain unknown. The model is calibrated to the current simulator's sampling and radio load, and ordinary TTL, battery-threshold and rolling rules receive the same information. Across 40 phase/harvest/variability cells and 5 declared death penalties, ordinary combinations match the exact non-prescient reference in service in 200/200 comparisons and in the priced objective up to numerical precision. The strict all-futures survival endpoint is infeasible at the declared tight-energy point, including for all-sparse operation. This is a risk-model boundary, not a claim that the monitoring mission or every physical implementation is infeasible.

The negative result closes the tested single-node stopping family. It neither supplies a new configuration algorithm nor establishes optimality for shared network scheduling or unseen task tables. Standard fallback remains a component of the execution system.

TABLE III

CONFIGURATION FALLBACK ACROSS TWO OUTAGE PHASES (3 SEEDS, `GENERIC-EXPIRY` AND `MAXCOV` FIXED). TTLS START AT DENSE APPLICATION. THE ANNOUNCED-VALIDITY ROW HAS TASK 1 INFORMATION; THE REMAINING ADAPTIVE/TTL POLICIES USE TASK 2 INFORMATION. THE THREE REPORTED ENERGY VARIANTS HAVE IDENTICAL CELL OUTCOMES, SO ONE ROW REPRESENTS THEM. ZERO OBSERVED DEATHS IS NOT A UNIVERSAL SURVIVAL GUARANTEE.

Local revert bound	Deaths (3 seeds)	Mean svc	Mean final SoC	Yellow delivered / total
<i>Phase A, overcast $\eta=0.012$, seeds 0–2 (upgrade 2 h, downgrade 6 h, outage 4–20 h)</i>				
never dense (blue only)	0	0.5752	0.00610	547/2016
clock guard (sunset 12 h)	20	0.5961	0.00329	849/2016
announced validity (6 h, Task 1)	0	0.5970	0.00610	849/2016
fixed TTL 4 h (revert 6 h)	0	0.5970	0.00610	849/2016
fixed TTL 8 h (revert 10 h)	0	0.5971	0.00552	849/2016
energy gate, open-loop bound	0	0.5771	0.00504	559/2016
<i>Phase B, overcast $\eta=0.012$, seeds 0–2 (upgrade 1 h, downgrade 8 h, outage 6–22 h)</i>				
never dense (blue only)	0	0.5399	0.00609	1168/3528
clock guard (sunset 12 h)	31	0.5982	0.00191	2020/3528
announced validity (8 h, Task 1)	0	0.5994	0.00609	2020/3528
fixed TTL 4 h (revert 5 h)	0	0.5928	0.00606	1916/3528
fixed TTL 8 h (revert 9 h)	0	0.5996	0.00606	2020/3528
energy gate, open-loop bound	0	0.5431	0.00416	1194/3528

D. What the field can attribute when the table arrives

In Episode I the delayed mission table reaches the gateway at $t_a=20$ h, by which point 2136 yellow obligations are past deadline. Offline time-aware attribution labels all 2136 correctly (S_{time} 588, S_{cap} 327, S_{access} 650, S_{energy} 571). Using only evidence the gateway lawfully holds by t_a labels 1138/2136 (53.3%)—all geometric S_{time} (588), all in-time-not-returned S_{cap} (327) and 223 provably late S_{access} —at 100% precision on the labelled subset; 998 (46.7%) remain unknown, whose offline truth is 571 never-sampled and 427 heard only after t_a (Table IV). “Never sampled” and “sampled but not yet heard” are inseparable without evidence the gateway does not have, and must stay unknown, which is the original G5 constraint. Compile-time geometry, needing no sample, emits 602 conditional no-return certificates of which 588 are true S_{time} , the 14 recovery-edge cases being excluded rather than certified; an optimistic-energy capacity certificate is impure (552 correct S_{cap} , 571 actually S_{energy} , 193 false positives), so capacity is left to the gateway. The design consequence is that expiry under a known local deadline can proceed while the remote cause remains unresolved; configuration fallback separately requires prior authority.

E. Enforcement placement determines survival

A centre-side feasibility *envelope*—a pure filter over any planner using only confirmed configs, hearing counts, the authorised requirement and the clock—removes the command storm but cannot save the tightest seed, because a dusk downgrade must cross the backhaul and a Class A window it misses. The same “never stay dense overnight” rule is a late, refusible suggestion at the centre and an enforceable local clamp at the node (Table V). Naive centre compliance kills 13 of 14 nodes on all three seeds (39 total) at service 0.162; adding the node-local guard removes all deaths at unchanged service; with the guard present the envelope additionally cuts refused admission attempts to 222. A zero-command pre-provisioned rhythm reaches 0.360 service, matching every centre-compiled arm

TABLE IV

ARRIVAL-TIME ATTRIBUTION OF THE 2136 UPGRADED OBLIGATIONS ALREADY MISSED WHEN THEIR TABLE REACHES THE GATEWAY (EPISODE I). THE ONLINE PROJECTOR IS EXACT BUT PARTIAL; 427 MATCHING SAMPLES ARE FIRST HEARD AFTER t_a AND 650 AFTER THEIR DEADLINE.

Segment	Offline truth	Online at t_a
S_{time} (no return slot; geometry)	588	588
S_{cap} (heard by deadline, not returned)	327	327
S_{access} (heard after deadline)	650	223
S_{energy} (no qualified sample)	571	— (unknown)
U (not separable online)	—	998
Total	2136	2136
Labelled coverage	100%	53.3%
Precision on labelled subset	1.000	1.000

for these static missions—a scoped negative result on centre sampling control for pre-provisioned tasks. The ordinary local controller is therefore part of the baseline for any further runtime or Agent comparison.

F. Formative real-agent study

We ran deepseek-flash live over the Episode I outage: three seeds each of an unaided agent (A0), the agent with an ordinary pending-command checklist (A0s) and a certificate agent (A1), plus two healthy-link controls, with a real model call at each 30-minute decision, 99 per trace and 1089 decisions in total; all arms share harness, legal observation, tools, history and action space. The study motivates and stress-tests the interface and is not used to establish the deterministic mechanisms.

Service and survival, noisy at $n=3$. A1 attains mean service 0.358, level with the 0.360 deterministic safe floor, with 4 deaths; A0 0.296 with 35 deaths and A0s 0.222 with 39; the healthy-link controls give A0 0.275/13 and A1 0.511/0. Per-seed service overlaps, so we draw no service significance. The live agent loop was not wired to the local guard, so the four

TABLE V

DETERMINISTIC EPISODE I PLACEMENT STUDY, NO LLM. THE LOCAL GUARD REMOVES ALL PERMANENT DEATHS IN THE TESTED SEEDS AT UNCHANGED SERVICE; THE CENTRE ENVELOPE ALONE CANNOT. REFUSED COMMANDS ARE REJECTED AT ADMISSION AND NEVER TRANSMITTED, WHICH IS COMMAND HYGIENE RATHER THAN BILLING SAVING. POINT-SPECIFIC EMPIRICAL RESULTS, NOT A SURVIVAL INVARIANT OR OPTIMAL BOUND.

Configuration	Mean svc	Deaths	Cmds sent	Refused
comply (naive, unprotected)	0.162	39	160	2488
dayfeed-c (centre day/night rule)	0.370	12	308	1256
envelope(comply), centre only	0.359	12	308	298
comply + local guard	0.366	0	1424	3364
dayfeed-c + local guard	0.368	0	258	1156
envelope(comply) + local guard	0.359	0	282	222
pure-local (pre-provisioned, 0 cmds)	0.360	0	0	0

A1 seed-1 deaths are not cleared by the deterministic result of §VII-E.

Call accounting. The 1089 decisions are not 1089 successful calls: 52 ended in a parse-failure hold from `max_tokens` cutting the JSON mid-string. Per-arm parse-failure rates are A0 7.1% (28/396), A0s 6.4% (19/297) and A1 1.3% (5/396), so arm differences mix output-format reliability with any semantic effect.

The interface fault, symmetrically scored. The observation’s link field carries only LoRa access receipts (`n_heard`) and no backhaul state. In the forced outage window A0/A0s assert “link up, commands can apply” on 36/32 decisions, and A1 on 1 under a token-aware statement check—over-optimistic, since the primary is down throughout, and accompanying the ineffective dense orders and deaths. In the opposite direction the v4 certificate asserted “control plane not delivering” whenever fewer than all nodes were heard, firing at five healthy-link times when the primary was in fact usable, twice after 13 of 14 nodes had already confirmed dense. Statements are checked in both outage and healthy conditions, independently of whether the decision issued a command. Offline replay of an unknown-aware v5 projector on identical snapshots issues zero unreachable assertions without a network-side field, turning all five healthy false alarms into unconfirmed, and splits 528 blanket night kills into 189 genuinely energy-infeasible and 339 advisory, every near-dawn 04:30–06:00 case being advisory. The finding is that a missing backhaul-state signal causes bidirectional errors; isolating any independent increment of the projector requires a matched-capability arm receiving the same constants and heuristics without the projector, end-to-end v5 runs with an explicit backhaul-state field, and cross-model replication, all future work.

VIII. RELATED WORK

LLMs in emergency and network control. TopoLLM applies adaptive tool learning to post-disaster topology planning [9], and later systems let LLMs schedule UAV paths and bandwidth, shape RL rewards or orchestrate recovery tools, including intent objects, budget and time constraints, candidate-plan checks, local repair and documented network-management APIs [10], [11]. We claim no novelty for natural

language to constraints, nor for tool orchestration. Our comparison concerns a different execution setting: persistent field state, Class A actuation and a separately interrupted return path. We use those works to position the controller interface and do not claim to have reproduced or refuted their original systems.

Scheduling, freshness, energy. The closest state-model precedent is an LLM in-context data-collection scheduler with bounded per-sensor queues, exogenous arrivals, finite service opportunities and feedback [12]; AoI and energy-harvesting control, including instantaneous-battery threshold policies and partially observed harvesting, address resource pieces [13], [14], [4]; overbooking and contact-aware forwarding for space links [15] and network-calculus bounds [16] are complementary deterministic characterisations. Joint sampling, transmission and energy optimisation is established and not claimed here; our distinction is persistent *remote* configuration with Class A apply latency, a separated access versus primary/backup return path, and termination under a control outage. Consistent network update research has long studied per-update validity, dependencies and safe transitions [17]; we rely on that lesson and add the disconnected, energy-constrained edge case in which an installed update’s *own data* become undeliverable before a counter-update can arrive.

Packet discard, DTN expiry, contacts. AoI management pre-empts or drops old updates [5], [18]; deadline-aware buffer discard has been coupled with adaptive ARQ for video [19]; DTN bundle protocols attach creation-time lifetimes and use TTL or encounter drop [6], [20], [7], and custody transfer [21] moves retention responsibility as a forward hand-off, since migrated out of the BPv7 core. Rural LoRa–LEO systems schedule around estimated satellite contacts and exploit spatial gateway diversity [22]. Our record disposition is BPv7-style expiry and we demonstrate equivalence on the tested seeds; ordinary TTL and battery protection also cover our evaluated configuration-stopping family. The measured system contribution is the placement and end-to-end effect of these primitives in the declared monitoring model. Custody transfer is a substantive alternative to central-ACK retention and must remain a comparator when feedback and storage responsibilities are changed.

IX. DISCUSSION, LIMITATIONS, AND THREATS TO VALIDITY

What the results establish. Source expiry, local protection and partial attribution have separate evidence. Their common lesson is to place decisions where the required evidence and ability to act survive disconnection. The strongest positive result concerns source retention; configuration stopping is a boundary study showing where ordinary mechanisms suffice. A better scheduler would not invalidate the observed retention or unreachable-downgrade failures.

Composition and generalisation. A complete execution runtime must be compared against an ordinary stack with the same expiry, local protection, configuration shadow, observations and planning tools. Component effects cannot be summed to infer an independent runtime gain. The present results cover a declared deployment and controlled episodes; outage phase, coupled access/backhaul failures and energy conditions delimit transfer. Firmware support for local fallback and expiry metadata, their signalling/storage cost, and hardware execution have not been established by the simulation results.

Feedback and retention. Central acknowledgement is synchronous and airtime-free in the model, equally favouring the compared queue rules. With an uplink-only backup, actual central delivery need not immediately become source knowledge. A deployed design must distinguish a local access receipt, durable gateway custody and central delivery, and account for any release notification sent through Class A windows. Standard custody transfer, cumulative acknowledgements and source expiry are therefore essential ordinary comparators. Joint allocation of finite feedback and configuration capacity remains outside the evaluated model.

Mission and energy scope. During Episode I, the source assigns expiry using the locally known pre-change cadence; central scoring still uses the actual mission. Multiple overlapping obligations may require retaining a record until its last valid use. Harvesting is synthetic and brownout is absorbing in the main configuration. The exact stopping reference has a public task table and a declared quantised energy model; its optimality statement does not transfer to an unknown mission or a complete communication network.

Agent evidence and remaining interface question. The live study uses one model and has both output-format and expert-information confounds. Its supported result is the interface failure and the projector's offline correction. A matched-capability end-to-end comparison must share local protection, evidence, deterministic tools and parsing budgets. The relevant outcome is whether an action-aware interface reduces harmful or ineffective execution while retaining beneficial changes, along with independently checked completion statements. Cross-model replication follows an isolated effect; more calls alone cannot establish it.

Provenance. Standard clauses were checked against the repository's source copies; one copy chain uses a document-sharing host because the official endpoint was unavailable. The instance parameters distinguish published capabilities from research choices. Cited systems' implementations were not

executed, and no reproduction of their reported performance is claimed.

X. CONCLUSION

Intermittent backhaul can remove control while installed communication state continues consuming resources. In the studied pre-disaster monitoring network, standard source expiry improves on-time service by 4.04 percentage points and outage delivery by $2.45\times$; local protection acts where a later central downgrade may be unreachable. Ordinary rules cover the evaluated configuration-stopping problem. The resulting design principle is to align retention and fallback with locally available evidence, authority and enforceable actuation, while keeping end-to-end completion distinct from intermediate receipts. An Agent can use this execution interface, but its independent benefit requires a matched ordinary-stack comparison. The present evidence supports the communication mechanisms and their boundaries, without making planner intelligence a prerequisite for correct local action.

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